

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

# Chapter 19: Constructions (Circles)

Topic-wise short notes with solved examples

This chapter covers the ruler-and-compass constructions connected with circles: tangents to a circle, and the circumscribed and inscribed circles of a triangle and of a regular hexagon.

## 1 Rules for every construction

- Use only a ruler and a pair of compasses, with a sharp pencil.
- Draw a rough sketch first and mark the given measurements on it.
- Leave all construction arcs and lines visible; they earn marks.
- Finish by measuring and writing down whatever the question asks for.
- The two basic constructions used again and again are the **perpendicular bisector** of a line segment and the **bisector of an angle**.

## 2 Tangent at a point on the circle

- **Given:** a circle with centre  $O$  and a point  $P$  on it.
- Step 1: join  $OP$ .
- Step 2: at  $P$ , construct a line  $APB$  perpendicular to  $OP$ .
- $APB$  is the required tangent.
- **Why it works:** the tangent at any point is perpendicular to the radius through that point.

## 3 Tangents from an external point

- **Given:** a circle with centre  $O$  and a point  $P$  outside it.
- Step 1: join  $OP$  and draw its perpendicular bisector to find the mid-point  $M$  of  $OP$ .
- Step 2: with centre  $M$  and radius  $MO$ , draw a circle. Let it cut the given circle at  $A$  and  $B$ .
- Step 3: join  $PA$  and  $PB$ . These are the two tangents.
- **Why it works:**  $OP$  is a diameter of the second circle, so  $\angle OAP = 90^\circ$  (angle in a semicircle). Thus  $PA$  is perpendicular to the radius  $OA$ .

**Example 3.1** A circle has radius 3 cm and a point  $P$  is 5 cm from its centre. Write the steps to construct the tangents from  $P$ , and calculate their length.

**SOLUTION**

Step 1: draw a circle with centre  $O$  and radius 3 cm. Mark  $P$  with  $OP = 5$  cm.

Step 2: bisect  $OP$  at  $M$ . With centre  $M$  and radius  $MO = 2.5$  cm draw a circle cutting the first circle at  $A$  and  $B$ .

Step 3: join  $PA$  and  $PB$ .

Length:  $PA = \sqrt{OP^2 - OA^2} = \sqrt{25 - 9}$

$PA = PB = 4$  cm (measure to confirm).

**REMEMBER**

Always check a tangent construction by calculation: the measured length should match  $\sqrt{d^2 - r^2}$ .

#### 4 Circumscribed circle of a triangle

- The **circumcircle** passes through all three vertices. Its centre is the **circumcentre** and its radius the **circumradius**.
- Step 1: construct the triangle.
- Step 2: draw the perpendicular bisectors of any two sides. Let them meet at  $O$ .
- Step 3: with centre  $O$  and radius  $OA$ , draw a circle. It passes through  $A$ ,  $B$  and  $C$ .
- Why it works:** every point on the perpendicular bisector of a side is equidistant from its two end points, so  $OA = OB = OC$ .

| Triangle      | Position of the circumcentre    |
|---------------|---------------------------------|
| Acute-angled  | Inside the triangle             |
| Right-angled  | The mid-point of the hypotenuse |
| Obtuse-angled | Outside the triangle            |

**Example 4.1** A right-angled triangle has sides 6 cm, 8 cm and 10 cm. Where is its circumcentre, and what is its circumradius?

**SOLUTION**

The circumcentre of a right-angled triangle is the mid-point of the hypotenuse.

**It is the mid-point of the 10 cm side, and the circumradius is 5 cm.**

#### 5 Inscribed circle of a triangle

- The **incircle** touches all three sides. Its centre is the **incentre** and its radius the **inradius**.
- Step 1: construct the triangle.
- Step 2: draw the bisectors of any two angles. Let them meet at  $I$ .

- Step 3: from  $I$  draw a perpendicular  $ID$  to any one side.
- Step 4: with centre  $I$  and radius  $ID$ , draw a circle. It touches all three sides.
- **Why it works:** every point on the bisector of an angle is equidistant from the arms of the angle, so  $I$  is equidistant from the three sides.
- The incentre always lies inside the triangle.

**Example 5.1** An equilateral triangle has side 6 cm. Calculate its inradius and its circumradius.

**SOLUTION**

Height =  $\frac{\sqrt{3}}{2} \times 6 = 3\sqrt{3}$  cm. In an equilateral triangle the incentre and circumcentre coincide at the centroid, which divides the height in the ratio 2 : 1.

Circumradius =  $\frac{2}{3} \times 3\sqrt{3} = 2\sqrt{3}$  cm and inradius =  $\frac{1}{3} \times 3\sqrt{3} = \sqrt{3}$  cm

**Inradius  $\approx 1.73$  cm; circumradius  $\approx 3.46$  cm**

## 6 Circles and a regular hexagon

- A regular hexagon has six equal sides and each interior angle is  $120^\circ$ .
- **To construct a regular hexagon of side  $a$ :** draw a circle of radius  $a$ , and with the same radius step off six arcs round the circle. Join the points in order.
- **Circumscribed circle** of a regular hexagon: draw the perpendicular bisectors of any two sides; they meet at the centre  $O$ . The radius is  $OA$ , and it **equals the side** of the hexagon.
- **Inscribed circle** of a regular hexagon: use the same centre  $O$ . Draw the perpendicular from  $O$  to any side; its length is the radius, equal to  $\frac{\sqrt{3}}{2}a$ .

**Example 6.1** A regular hexagon has side 4 cm. Find the radius of its circumscribed circle and of its inscribed circle.

**SOLUTION**

Circumradius = side = 4 cm

Inradius =  $\frac{\sqrt{3}}{2} \times 4 = 2\sqrt{3}$  cm

**4 cm and about 3.46 cm**

## 7 Common mistakes

- Using perpendicular bisectors for the incircle, or angle bisectors for the circumcircle.
- Taking the radius of the incircle as the distance from  $I$  to a vertex. It is the **perpendicular** distance to a side.
- Drawing a tangent by eye instead of constructing it.
- Rubbing out the arcs.